1st Place: Aspect 6 – Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms

- **Analysis**: This is the most central competency for this problem. Students must accurately understand the formula $D = \frac{dv}{60n}$ as a given "formal system" or mathematical rule and reason within this system. They need to master basic rules of algebraic manipulation (e.g., fraction operations, variable substitution) to deduce the change.
- **Manifestation in the Question**: Solving the problem relies entirely on understanding the structure of this specific formula. Students need to see that the relationship between D and n in the formula is inversely proportional (since n is in the denominator). They need to apply the algebraic rule that "if the denominator of a fraction is doubled while the numerator remains unchanged, the value of the fraction is halved."

2nd Place: Aspect 7 – Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- **Analysis**: This problem deals not with numbers or graphs, but purely with **algebraic expressions** and the relationships between variables within them.
- **Manifestation in the Question**: Students need to directly "manipulate" the algebraic formula $D = \frac{dv}{60n}$. This includes identifying variables, understanding the operational structure (division), and performing operations at the symbolic level (e.g., substituting n with $2n$ and then comparing the new expression with the original).

3rd Place: Aspect 8 – Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- **Analysis**: The problem requires to "Describe precisely how D changes." This demands not only the correct qualitative conclusion ("halves") but also a precise quantitative description or a clear reasoning process.
- **Manifestation in the Question**: A complete response should articulate the process of change. For example: "When n becomes $2n$, and d and v remain unchanged, the new drip rate $D' = \frac{dv}{60*(2n)} = \frac{1}{2} * \frac{dv}{60n} = \frac{1}{2}D$. Therefore, the drip rate D is halved." This shows the complete intermediate results from substitution to simplification, and then to comparison, leading to the final conclusion.

4th Place: Aspect 12 – Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- **Analysis**: Although the reasoning chain might be short, it still follows a logical sequence of steps.
- **Manifestation in the Question**: A typical reasoning procedure is: 1) Write down the original formula; 2) Substitute n with $2n$ to write the new expression; 3) Simplify the new expression and compare it with the original formula; 4) Draw the exact conclusion about how D changes. This is a concise but complete multi-step procedure.

5th Place: Aspect 13 – Use an understanding of the context to guide or expedite the mathematical solving process

- **Analysis**: The problem is rooted in a medical context (infusion drip rate). Although the mathematical derivation is purely algebraic, understanding the context helps verify the reasonableness of the result.
- **Manifestation in the Question**: Understanding that "doubling the infusion time" intuitively means "the drip rate should slow down" can serve as a sanity check. If the algebraic derivation suggested the drip rate doubled, students might sense an error. The context provides a real-world anchor for the abstract algebraic reasoning.

6th Place: Aspect 1 – Perform a simple calculation

- **Analysis**: The calculation itself is very simple, primarily involving basic fraction operations (dividing by 2).
- **Manifestation in the Question**: Deriving $D' = \frac{dv}{120n}$ or $\frac{1}{2}D$ involves basic arithmetic calculation.

第1位：方面 6 - 理解并运用基于定义、规则和形式系统的构造

- **分析**：这是本题最核心的能力。学生必须准确理解公式 $D = \frac{dv}{60n}$ 作为一个给定的“形式系统”或数学规则，并在这个系统内进行推理。他们需要掌握代数操作的基本规则（如分数运算、变量代换）来推演变化。
- **题目体现**：解决问题完全依赖于对这个特定公式结构的理解。学生需要看到 D 与 n 在公式中的关系是反比关系（因为 n 在分母上）。他们需要运用“如果一个分数分母加倍而分子不变，则分数值减半”这一代数规则。

第2位：方面 7 - 处理数字、图形与统计信息、代数表达式和方程，以及几何表示

- **分析**：本题处理的不是数字或图形，而是纯粹的**代数表达式**和其中的变量关系。
- **题目体现**：学生需要直接“处理”代数公式 $D = \frac{dv}{60n}$。这包括识别变量、理解运算结构（除法），并进行符号层面的操作（例如，将 n 替换为 $2n$，然后比较新表达式与原表达式）。

第3位：方面 8 - 阐明解决方案，展示和/或总结、呈现中间数学结果

- **分析**：题目要求“Describe precisely how D changes”。这不仅要求得出正确的定性结论（“减半”），还要求给出精确的定量描述或清晰的推理过程。
- **题目体现**：一个完整的回答应阐明变化过程。例如：“当 n 变为 $2n$，而 d 和 v 不变时，新的滴速 $D' = \frac{dv}{60*(2n)} = \frac{1}{2} * \frac{dv}{60n} = \frac{1}{2}D$。因此，滴速 D 减半。” 这展示了从代入到简化，再到比较的完整中间结果和最终结论。

第4位：方面 12 - 使用多步骤流程得出数学解答或结论

- **分析**：虽然推理链条可能很短，但它依然遵循一个逻辑步骤序列。
- **题目体现**：典型的推理步骤是：1) 写出原公式；2) 将 n 替换为 $2n$，写出新表达式；3) 简化新表达式，将其与原公式比较；4) 得出关于 D 变化的确切结论。这是一个简洁但完整的多步流程。

第5位：方面 13 - 利用对情境的理解来指导或加速求解过程

- **分析**：问题植根于医疗情境（输液滴速）。尽管数学推导是纯代数的，但理解情境有助于验证结果的合理性。
- **题目体现**：理解“输液时间加倍”在直觉上意味着“滴速应该变慢”，这可以作为一个合理性检查。如果代数推导得出滴速翻倍，学生可能会意识到错误。情境为抽象的代数推理提供了现实锚点。

第6位：方面 1 - 执行简单计算

- **分析**：计算本身非常简单，主要是分数的基本运算（除以2）。
- **题目体现**：在推导 $D' = \frac{dv}{120n}$ 或 $\frac{1}{2}D$ 时，涉及了基本的算术计算。